Name: Talha Rusman
NU ID: 23K-0065
Instructor: Sir Omar Qureshi

ASSIGNMENT # 1

Reflective Summary

Introduction and Reflection

The lecture about Variational Autoencoders (VAEs) helped me see how a basic autoencoder can be developed into a model that generates new data. A traditional autoencoder encodes an input into a fixed latent representation and then tries to reconstruct the original input. However, its latent space may not be organized in a way that supports reliable random sampling, which can lead to unrealistic generated results. A VAE improves this by learning a probability distribution in the latent space, creating a smoother and more suitable space for producing new examples.

How a VAE Works

Rather than returning a single latent vector, the encoder in a VAE estimates the mean (μ) and standard deviation (σ) of a distribution, written as $q\varphi(z|x)$. A latent sample is then drawn from this distribution and passed to the decoder. This process can be expressed as $z = \mu + \sigma \odot \varepsilon$, where ε follows a standard normal distribution, $\varepsilon \sim N(0, I)$. This method is known as the reparameterization trick. It places the randomness in ε while keeping μ and σ differentiable, so the model can update these parameters through backpropagation.

Loss Functions in a VAE

The reconstruction loss evaluates how closely the decoder's output ($\hat{x}$) matches the original input (x). For continuous-valued data, Mean Squared Error (MSE) is one possible choice. For binary or normalized data, Binary Cross-Entropy (BCE) may be appropriate. In probabilistic terms, the reconstruction component can also be expressed as the negative expected log-likelihood: $L_reconstruction = E_q\varphi(z|x)[-\log p\theta(x|z)]$.

Another key part of the objective is Kullback–Leibler (KL) divergence. It measures the difference between the encoder's distribution $q\varphi(z|x)$ and the standard normal prior $p(z) = N(0, I)$. For a Gaussian posterior, the KL term is calculated as $L_KL = -\frac{1}{2} \Sigma [1 + \log(\sigma^2) - \mu^2 - \sigma^2]$. This penalty encourages the learned latent distributions to stay close to the standard normal prior, helping the latent space remain smooth and continuous. The total VAE objective combines both parts: $L_VAE = L_reconstruction + L_KL$.

Difference Between VAE and β-VAE

The key distinction between a standard VAE and a β-VAE is the coefficient applied to the KL-divergence term. In a regular VAE, β equals 1, so the loss consists of the reconstruction term plus the KL penalty. A β-VAE generally uses a value of β greater than 1, increasing the emphasis on latent-space regularization: $L_\beta\text{-}VAE = L_reconstruction + \beta L_KL$.

A larger β can encourage the latent dimensions to capture more separate, or disentangled, factors of variation, making the representation easier to understand. However, applying too much regularization may harm reconstruction accuracy and produce outputs that look excessively smooth or blurry. In this way, a standard VAE focuses on balancing reconstruction with regularization, whereas a β-VAE gives greater priority to learning a structured and more disentangled latent space.

Conclusion

Overall, the lecture helped me understand that VAEs are not simply tools for compressing data. They learn a probabilistic latent space that can be sampled to create new examples. The ideas I found most important were the reparameterization trick, the two main loss components—reconstruction and KL divergence—and the role of β in a β-VAE. These concepts clarify how VAEs support data generation, representation learning, and other machine-learning tasks.